MGU-UGP BSc (HONS) MICROBIOLOGY

III SEMESTER

DSC B - MG3DSCMBG202–TECHNIQUES INMICROBIOLOGY

(2024 ONWARDS)

Time: 1.5 Hours **Max. marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions

Answer any 10 questions.

Each question carries 2 marks.

1. Define sterilisation. [K][1]
2. Write down the application of HEPA filters in microbiology. [U][1]
3. How are thermolabile substances sterilised? [K][1]
4. Explain the mechanism of action of ethylene oxide in sterilisation. [U][1]
5. Name two sources of agar. [K][2]
6. Explain the role of transport media in a microbiology laboratory. [U][2]
7. Discuss enriched media with examples. [U][2]
8. List the different methods used for the isolation of microorganisms. [K][3]
9. What is Cryopreservation? [K][3]
10. Explain negative staining with an example.[U][4]
11. What is Numerical Aperture? [K][4]
12. Distinguish between SEM and TEM?[U][4]

(10x2=20)

Part B

Short Essay Questions

Answer any 4 questions.

Each question carries 5 marks

13. Compare the role ofionisingand non-ionisingradiation in sterilisation.? [U][1]
14. Explain briefly about pasteurisation and its types. [U][1]
15. Give a detailed account of anaerobic culture media. [U][2]
16. Write a note on the maintenance &preservation of pure culture. [K][3]
17. Explain the pour plate method for separating and counting bacterial isolates in a water sample. [A][3]
18. Describe the principle and procedure of Gram staining. [K][4]

(4x5=20)

Part C

Long Essay Question

Answer any 1question.

Each question carries 10 marks.

19. Describe the general safety rules to be followed in a microbiology laboratory. [U][1]

20. Explain the principle and working of the fluorescent microscope and phase contrast microscope with a suitable diagram. [A][4]

(10x1=10)